

Laboratorio docker

Angel Gonzalez - angonpad@gmail.com

0. Preparación

Verificamos docker y si esta corriendo el daemon

```
~/Projects/docker-test 19:40
> docker version
Client:
 Version:           29.3.1
 API version:       1.54
 Go version:        go1.26.1-X:nodwarf5
 Git commit:        c2be9ccfc3
 Built:             Wed Mar 25 19:07:20 2026
 OS/Arch:           linux/amd64
 Context:           default

Server:
 Engine:
  Version:          29.3.1
  API version:      1.54 (minimum version 1.40)
  Go version:       go1.26.1-X:nodwarf5
  Git commit:       f78c987ad3
  Built:            Wed Mar 25 19:07:20 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.2
  GitCommit:        301b2dac98f15c27117da5c8af12118a041a31d9.m
 runc:
  Version:          1.4.1
  GitCommit:
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

```
Λ ~/Projects/docker-test 19:41
> docker info
Client:
Version:      29.3.1
Context:      default
Debug Mode:   false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  0.32.1
    Path:      /usr/lib/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:   5.1.1
    Path:      /usr/lib/docker/cli-plugins/docker-compose

Server:
Containers: 0
  Running: 0
  Paused: 0
  Stopped: 0
Images: 4
Server Version: 29.3.1
Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: true
  Native Overlay Diff: false
  userxattr: false
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
  /etc/cdi
  /var/run/cdi
Swarm: inactive
Runtimes: runc io.containerd.runc.v2
Default Runtime: runc
Init Binary: docker-init
containerd version: 301b2dac98f15c27117da5c8af12118a041a31d9.m
runc version:
init version: de40ad0
Security Options:
  seccomp
   Profile: builtin
  cgroupns
Kernel Version: 6.19.9-arch1-1
Operating System: Arch Linux
OSType: linux
Architecture: x86_64
CPUs: 4
Total Memory: 7.451GiB
Name: ArchLinux
ID: 9f682c00-0fc7-4c42-9736-3d3513bbda9f
Docker Root Dir: /var/lib/docker
Debug Mode: false
Experimental: false
Insecure Registries:
  ::1/128
  127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables
```

1. Docker image
 - 1.1. Descargar una imagen

```
~/Projects/docker-test 19:42
> docker pull nginx:alpine
alpine: Pulling from library/nginx
589002ba0eae: Already exists
8892f80f46a0: Pull complete
91d1c9c22f2c: Pull complete
cf1159c696ee: Pull complete
3f4ad4352d4f: Pull complete
c2bd5ab17727: Pull complete
4d9d41f3822d: Pull complete
3370263bc02a: Pull complete
Digest: sha256:e7257f1ef28ba17cf7c248cb8ccf6f0c6e0228ab9c315c152f9c203cd34cf6d1
Status: Downloaded newer image for nginx:alpine
docker.io/library/nginx:alpine
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
myapp:v1	1196fe9de3c9	8.44MB	0B	
mysql:8.0	34178d8aefda	783MB	0B	
nginx:alpine	d5030d429039	62.2MB	0B	
redis:6.2-alpine	fcccc1a5a8341	30.2MB	0B	

- 1.2. Construir una imagen desde Dockerfile

```
1 FROM alpine:3.20
2 WORKDIR /app
3 RUN echo "Hola desde Docker" > /app/mensaje.txt
4 CMD ["cat", "/app/mensaje.txt"]
```

NORMAL dockerfile utf-8 4:31

```

Λ ~/Projects/docker-test 19:44
> docker build -t myapp:v1 .
[+] Building 2.9s (7/7) FINISHED                                docker:default
=> [internal] load build definition from dockerfile              0.0s
=> => transferring dockerfile: 147B                             0.0s
=> [internal] load metadata for docker.io/library/alpine:3.20   1.5s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                   0.0s
=> [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372e 1.1s
=> => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372e 0.0s
=> => sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805 9.22kB / 9.22kB 0.0s
=> => sha256:b0cb30c51c47cdfde647364301758b14c335dea2fddc9490d4f007d67ecb2538 1.02kB / 1.02kB 0.0s
=> => sha256:cc9071bd161080c1a543f3023b7d0db905b497e6ae757fe078227803bc7e4dc8 611B / 611B 0.0s
=> => sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 3.63MB / 3.63MB 1.0s
=> => extracting sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 0.1s
=> [2/3] WORKDIR /app                                           0.0s
=> [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt      0.1s
=> exporting to image                                           0.1s
=> => exporting layers                                          0.1s
=> => writing image sha256:c41c3454dee3d58a4cf228cb3b92c9777f73bc475676d63ccfb12c2d648a8254 0.0s
=> => naming to docker.io/library/myapp:v1                     0.0s

```

```

Λ ~/Projects/docker-test 19:45
> docker images | grep myapp
WARNING: This output is designed for human readability. For machine-readability
myapp:v1          c41c3454dee3          7.8MB          0B

```

```

Λ ~/Projects/docker-test 19:45
> docker run --name test myapp:v1
Hola desde Docker

```

1.3. Etiquetar una imagen

```

Λ ~/Projects/docker-test 19:47
> docker images | grep myapp
WARNING: This output is designed for human readability. For machine-readability
myapp:latest      c41c3454dee3          7.8MB          0B          U
myapp:v1          c41c3454dee3          7.8MB          0B          U

```

1.4. Inspeccionar una imagen

```
STDIN
1  [
2    {
3      "Id": "sha256:c41c3454dee3d58a4cf228cb3b92c9777f73bc475676d63ccfb12c2d648a8254",
4      "RepoTags": [
5        "myapp:latest",
6        "myapp:v1"
7      ],
8      "RepoDigests": [],
9      "Comment": "buildkit.dockerfile.v0",
10     "Created": "2026-03-26T19:45:04.934081322-05:00",
11     "Config": {
12       "Env": [
13         "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
14       ],
15       "Cmd": [
16         "cat",
17         "/app/mensaje.txt"
18       ],
19       "WorkingDir": "/app",
20       "ArgsEscaped": true
21     },
22     "Architecture": "amd64",
23     "Os": "linux",
24     "Size": 7800292,
25     "GraphDriver": {
26       "Data": {
27         "LowerDir": "/var/lib/docker/overlay2/i2a4iziovqpnjxml6rou8ent/diff:/var/lib/dock
28         er/overlay2/6ed15dbdd7dc8f52ec5a307ad82812567812a31af49923b076879229ca3aade6/diff",
29         "MergedDir": "/var/lib/docker/overlay2/o0j0ossvpf3egu6mlxkzjycpg/merged",
30         "UpperDir": "/var/lib/docker/overlay2/o0j0ossvpf3egu6mlxkzjycpg/diff",
31         "WorkDir": "/var/lib/docker/overlay2/o0j0ossvpf3egu6mlxkzjycpg/work"
32       },
33       "Name": "overlay2"
34     },
35     "RootFS": {
36       "Type": "layers",
37       "Layers": [
38         "sha256:f5b2c0456bf1080bb3f0617295df9721aa4aa27a536bba8e301c3af8d530c9cb",
39         "sha256:a86077bb4ede60f8221b7f00d28f4614fd95df4117a2690929b4201a5a432f1f",
40         "sha256:641d927b3dacb14807fe74868ca314d736a4d0ba21706ad7a348049230cc4981"
41       ]
42     },
43     "Metadata": {
44       "LastTagTime": "2026-03-26T19:47:19.629634031-05:00"
45     }
46   }
47 ]
```

```
~/Projects/docker-test 19:50
> docker image inspect myapp:latest --format '{{.Id}}'
sha256:c41c3454dee3d58a4cf228cb3b92c9777f73bc475676d63ccfb12c2d648a8254
```

1.5. Guardar una imagen a tar

```
~/Projects/docker-test 19:51
> ls -lh
Permissions Size User Date Modified Name
-rw-r--r-- 110 pukiretsu 26 Mar 19:44 dockerfile
-rw----- 8.1M pukiretsu 26 Mar 19:51 myapp.tar
```

1.6. Cargar una imagen desde tar

```
~/Projects/docker-test 18:23
> docker rmi myapp:v1
Untagged: myapp:v1
```

```
~/Projects/docker-test 18:24
> docker load -i myapp.tar
Loaded image: myapp:latest

~/Projects/docker-test 18:24
> docker images | grep myapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
myapp:latest      c41c3454dee3      7.8MB      0B      U
```

1.7. Eliminar una imagen

```
~/Projects/docker-test 18:25
> docker rmi -f myapp:latest
Untagged: myapp:latest
Deleted: sha256:c41c3454dee3d58a4cf228cb3b92c9777f73bc475676d63ccfb12c2d648a8254
```

```
~/Projects/docker-test 18:25
> docker images | grep myapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
```

1.8. Limpiar imágenes no usadas

```
~/Projects/docker-test 19:51
> docker image prune -f
Deleted Images:
untagged: nginx@sha256:b3c656d55d7ad751196f21b7fd2e8d4da9cb430e32f646adcf92441b72f82b14
deleted: sha256:d4918ca78576a537caa7b0c043051c8efc1796de33fee8724ee0fff4a1cabed9
deleted: sha256:88af9f575d6a9fa7b5a16120f2a18f8113bbe9c8345586c40d360ba3042572a1
deleted: sha256:db26c4e59bebcfa9acae6b53341063c603cbbc911bc2e3bac6edc8bdf0fbc686
deleted: sha256:dfb0b3d16d5c315aacf955587e75cfa9d9e65d6744dce0dc107cad0e56bf10d6
deleted: sha256:583c66ba990448f304385e3bf32295d3431a57d9ae446924da19a6256e1d8de5
deleted: sha256:8cb0e9a8e468f16e92dfed125f82f9010a7c4a771ef9ec021806f297ae338814
deleted: sha256:f91420e68a07766231185cdc55b7da14ce47fae1bdba44f872859fc4d438817c
deleted: sha256:79d5db50f2457635365b7212e8f8911a9f3f3522f52b6cc01c81579f52ac87b9
deleted: sha256:256f393e029fa2063d8c93720da36a74a032bed3355a2bc3e313ad12f8bde9d1
deleted: sha256:1196fe9de3c9dddb4b5148f379f9185cf2702a30c9a8385cdec4c25006f3639

Total reclaimed space: 52.82MB
```

2. Docker container

2.1. Ejecutar un contenedor

```
~/Projects/docker-test 19:53
> docker run -d --name webl nginx:latest
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
ec781dee3f47: Pull complete
bb3d0aa29654: Pull complete
510ddf6557d6: Pull complete
cde7a05ae428: Pull complete
587e3d84dbb5: Pull complete
3189680c601f: Pull complete
5e815e07e569: Pull complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
395deb214f1fe650158311439d9d537576f93e2b6365d40c9613b8dfb90f7144
```

```
~/Projects/docker-test 19:53
> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
395deb214f1f   nginx:late "/docker-entrypoint..." 25 seconds ago Up 24 seconds 80/tcp       webl
```

2.2. Ejecutar contenedor con nombre y puertos

```
~/Projects/docker-test 19:54
> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
77275905799e   nginx:late "/docker-entrypoint..." 3 seconds ago  Up 3 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
395deb214f1f   nginx:late "/docker-entrypoint..." About a minute ago Up About a minute 80/tcp       webl
```

```
~/Projects/docker-test 19:55
> curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

2.3. Listar todos los contenedores

```
~/Projects/docker-test 19:55
> docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
77275905799e   nginx:late "/docker-entrypoint..." About a minute ago Up About a minute 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
395deb214f1f   nginx:late "/docker-entrypoint..." 2 minutes ago  Up 2 minutes  80/tcp       webl
b94dbe19f98d   myapp:v1   "cat /app/mensaje.txt" 9 minutes ago  Exited (0) 9 minutes ago                                test
```


2.4. Ver logs

```
> docker logs web2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/27 00:54:46 [notice] 1#1: using the "epoll" event method
2026/03/27 00:54:46 [notice] 1#1: nginx/1.29.7
2026/03/27 00:54:46 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/27 00:54:46 [notice] 1#1: OS: Linux 6.19.9-arch1-1
2026/03/27 00:54:46 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/03/27 00:54:46 [notice] 1#1: start worker processes
2026/03/27 00:54:46 [notice] 1#1: start worker process 29
2026/03/27 00:54:46 [notice] 1#1: start worker process 30
2026/03/27 00:54:46 [notice] 1#1: start worker process 31
2026/03/27 00:54:46 [notice] 1#1: start worker process 32
172.17.0.1 - - [27/Mar/2026:00:55:27 +0000] "\x16\x03\x01\x06\x11\x01\x00\x06" 400 157 "-" "-" "-"
172.17.0.1 - - [27/Mar/2026:00:55:33 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.19.0" "-"
```

2.5. Inspeccionar

```
STDIN
[
  {
    "Id": "77275905799e813bb995a8e01caacd328cbd43f55ce59d96c01b97b140fc54ee",
    "Created": "2026-03-27T00:54:46.187059697Z",
    "Path": "/docker-entrypoint.sh",
    "Args": [
      "nginx",
      "-g",
      "daemon off;"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 20587,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-03-27T00:54:46.220225187Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:0cf1d6af5ca72e2ca196afdbde26d96f141bd3dc14d70210707cf89032ea217",
    "ResolvConfPath": "/var/lib/docker/containers/77275905799e813bb995a8e01caacd328cbd43f55ce59d96c01b97b140fc54ee/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/77275905799e813bb995a8e01caacd328cbd43f55ce59d96c01b97b140fc54ee/hostname",
    "HostsPath": "/var/lib/docker/containers/77275905799e813bb995a8e01caacd328cbd43f55ce59d96c01b97b140fc54ee/hosts",
    "LogPath": "/var/lib/docker/containers/77275905799e813bb995a8e01caacd328cbd43f55ce59d96c01b97b140fc54ee/77275905799e813bb995a8e01caacd328cbd43f55ce59d96c01b97b140fc54ee-json.log",
    "Name": "/web2",
    "RestartCount": 0,
    "Driver": "overlay2",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": "",
      "LogConfig": {
        "Type": "json-file",
        "Config": {}
      },
      "NetworkMode": "bridge",

```

```
~/Projects/docker-test 19:57
> docker inspect web2 --format '{{.State.Status}}'
running
```

2.6. detener e iniciar un contenedor

```
~/Projects/docker-test 20:00
> docker stop web2
web2

~/Projects/docker-test 20:01
> docker start web2
web2

~/Projects/docker-test 20:01
> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
77275905799e   nginx:late "/docker-entrypoint..." 7 minutes ago  Up 3 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  web2
395deb214f1f   nginx:late "/docker-entrypoint..." 8 minutes ago  Up 8 minutes  80/tcp                               web1
```


2.7. Entrar en modo interactivo

```
~/Projects/docker-test 19:58
> docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
76eb174b37c3: Already exists
Digest: sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.9
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit
```

```
~/Projects/docker-test 20:00
> docker ps -a | grep alpine1
6f353be6a40f alpine:3.20 "sh" About a minute ago Exited (0) 44 seconds ago alpine1
```

2.8. Pausar y reanudar

```
~/Projects/docker-test 20:02
> docker inspect webl --format '{{.State.Paused}}'
true
```

```
~/Projects/docker-test 20:02
> docker unpause webl
webl
```

```
~/Projects/docker-test 20:03
> docker inspect webl --format '{{.State.Paused}}'
false
```

2.9. Eliminar contenedores

```
~/Projects/docker-test 18:13
> docker rm alpine1
alpine1
```

```
~/Projects/docker-test 18:14
> docker rm -f webl
webl
```

```
~/Projects/docker-test 18:15
> docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
77275905799e	nginx:latest	"/docker-entrypoint..."	3 weeks ago	Exited (0) 3 weeks ago		web2
b94dbe19f98d	myapp:v1	"cat /app/mensaje.txt"	3 weeks ago	Exited (0) 3 weeks ago		test

3. Volúmenes

3.1. Crear volumen

```
~/Projects/docker-test 18:25
> docker volume create nginx_data
nginx_data
```

```
~/Projects/docker-test 18:29
> docker volume ls
DRIVER      VOLUME NAME
local       6a451c6325ec6137c35e63df894d27be275458e53522c9ade26d5d006384b8b2
local       08f7d0c49e3013a947caccb94ca5953f54b3768ce2aa53213cd457cb7bbb634f
local       0151433702da3a1f4b62186bcecf811596e81fab09708f127421d7b6c121c02
local       api-tasks_composer_cache
local       api-tasks_dbdata
local       b79955d7c457ddbcb7fc17d201357f0738eea3b66b019383a9108982a794469d
local       d9de040f6d841096936cc36e1b3d533953e3db6c57f63503b7d23574bb6f2629
local       db90665896a1dc9517f80b8203a99e69971c4bff589f3032d0c81383426a6c12
local       magento-docker_m2-db-data
local       nginx_data
```

3.2. Inspeccionar volumen

```
~/Projects/docker-test 18:29
> docker volume inspect nginx_data
[
  {
    "CreatedAt": "2026-04-20T18:29:25-05:00",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/nginx_data/_data",
    "Name": "nginx_data",
    "Options": null,
    "Scope": "local"
  }
]
```

3.3. Usar volumen en un contenedor

```
~/Projects/docker-test 19:06
> docker run -d --name webvol -p 8081:80 -v nginx_data:/usr/share/nginx/html nginx:latest
bc748dafb8f5473f115984986316aca6501e7087779eb37c00ce4268e282466c
```

```
~/Projects/docker-test 19:12
> cat index.html
```

	File: index.html
1	Hola volumen

```
~/Projects/docker-test 19:12
> docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 22B (transferred 2.05kB) to webvol:/usr/share/nginx/html/index.html
```

```
~/Projects/docker-test 19:13
> curl http://localhost:8081
<h1>Hola volumen</h1>
```

3.4. eliminar volumen

```

~ /Projects/docker-test 19:15
> docker rm -f webvol
webvol

~ /Projects/docker-test 19:15
> docker volume rm nginx_data
nginx_data

~ /Projects/docker-test 19:20
> docker volume ls
DRIVER          VOLUME NAME
local           6a451c6325ec6137c35e63df894d27be275458e53522c9ade26d5d006384b8b2
local           08f7d0c49e3013a947caccb94ca5953f54b3768ce2aa53213cd457cb7bbb634f
local           0151433702da3a1f4b62186bcecf811596e81fab09708f127421d7b6c121c02
local           api-tasks_composer_cache
local           api-tasks_dbdata
local           b79955d7c457ddbcb7fc17d201357f0738eea3b66b019383a9108982a794469d
local           d9de040f6d841096936cc36e1b3d533953e3db6c57f63503b7d23574bb6f2629
local           db90665896a1dc9517f80b8203a99e69971c4bfff589f3032d0c81383426a6c12
local           magento-docker_m2-db-data

~ /Projects/docker-test 19:20
>

```

4. Redes

4.1. Listar redes

```

A ~/Projects/docker-test 19:21
> docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
ed4bf463e754	api-tasks_laravel_network	bridge	local
c61d6599a6fb	bridge	bridge	local
2a3ea71b2658	host	host	local
ef391a72e744	mariadb_coursera_default	bridge	local
3ee7ce3b7b58	none	null	local

4.2. Crear red bridge propia

```
~/Projects/docker-test 19:21
> docker network create my_network
008a96300f9fa12cfe8d46ce7f8be69667b3afd7c61fc83fae003722d6f8b6dd

~/Projects/docker-test 19:22
> docker network ls
NETWORK ID          NAME                                DRIVER              SCOPE
ed4bf463e754        api-tasks_laravel_network         bridge              local
c61d6599a6fb        bridge                             bridge              local
2a3ea71b2658        host                              host                local
ef391a72e744        mariadb_coursera_default          bridge              local
008a96300f9f        my_network                        bridge              local
3ee7ce3b7b58        none                              null                local
```

4.3. Inspeccionar red

```
STDIN
1  [
2    {
3      "Name": "my_network",
4      "Id": "008a96300f9fa12cfe8d46ce7f8be69667b3afd7c61fc83fae003722d6f8b6dd",
5      "Created": "2026-04-20T19:22:34.777348904-05:00",
6      "Scope": "local",
7      "Driver": "bridge",
8      "EnableIPv4": true,
9      "EnableIPv6": false,
10     "IPAM": {
11       "Driver": "default",
12       "Options": {},
13       "Config": [
14         {
15           "Subnet": "172.20.0.0/16",
16           "Gateway": "172.20.0.1"
17         }
18       ]
19     },
20     "Internal": false,
21     "Attachable": false,
22     "Ingress": false,
23     "ConfigFrom": {
24       "Network": ""
25     },
26     "ConfigOnly": false,
27     "Options": {},
28     "Labels": {},
29     "Containers": {},
30     "Status": {
31       "IPAM": {
32         "Subnets": {
33           "172.20.0.0/16": {
34             "IPsInUse": 3,
35             "DynamicIPsAvailable": 65533
36           }
37         }
38       }
39     }
40   }
41 ]
```

4.4. Conectar contenedores a una red

```
Λ ~/Projects/docker-test 19:26
> docker run -d --name caja1 --network my_network alpine:3.2 sleep 300
Unable to find image 'alpine:3.2' locally
3.2: Pulling from library/alpine
95f5ecd24e43: Pull complete
Digest: sha256:e9a2035f9d0d7cee1cdd445f5bfa0c5c646455ee26f14565dce23cf2d2de7570
Status: Downloaded newer image for alpine:3.2
6d60b3559784019d357da1769aa7e0c1db2b00443f4ab8bcb0d36dab51b549e9

Λ ~/Projects/docker-test 19:27
> docker run -d --name caja2 --network my_network alpine:3.2 sleep 300
5dcdc7daaad807498a214d0d029217b88a0eed557bc77d01c90048e0d6afec17

Λ ~/Projects/docker-test 19:27
> docker exec caja1 ping -c 2 caja2
PING caja2 (172.20.0.3): 56 data bytes
64 bytes from 172.20.0.3: seq=0 ttl=64 time=0.062 ms
64 bytes from 172.20.0.3: seq=1 ttl=64 time=0.076 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.062/0.069/0.076 ms
```

4.5. Desconectar un contenedor de la red

```
Λ ~/Projects/docker-test 19:29
> docker network disconnect my_network caja2

Λ ~/Projects/docker-test 19:29
> docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'
```

5. Docker compose

5.1. Crear archivo compose

```
Λ ~/Projects/docker-test 19:38
> cat compose.yml
```

	File: compose.yml
1	services:
2	web:
3	image: nginx:latest
4	ports:
5	- "8082:80"

5.2. Levantar servicios

```
~/Projects/docker-test 19:38
> docker compose up -d
[+] up 2/2
✓ Network docker-test_default Created
✓ Container docker-test-web-1 Started

~/Projects/docker-test 19:40
> docker compose ps
NAME                IMAGE            COMMAND                  SERVICE    CREATED   STATUS    PORTS
docker-test-web-1   nginx:latest     "/docker-entrypoint..." web         5 seconds ago Up 5 seconds 0.0.0.0:8082→80/tcp, [::]:8082→80/tcp

~/Projects/docker-test 19:40
> curl http://localhost:8082
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

5.3. Ver logs

```
~/Projects/docker-test 19:40
> docker compose logs
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/04/21 00:40:35 [notice] 1#1: using the "epoll" event method
web-1 | 2026/04/21 00:40:35 [notice] 1#1: nginx/1.29.7
web-1 | 2026/04/21 00:40:35 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1 | 2026/04/21 00:40:35 [notice] 1#1: OS: Linux 6.19.11-arch1-1
web-1 | 2026/04/21 00:40:35 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
web-1 | 2026/04/21 00:40:35 [notice] 1#1: start worker processes
web-1 | 2026/04/21 00:40:35 [notice] 1#1: start worker process 29
web-1 | 2026/04/21 00:40:35 [notice] 1#1: start worker process 30
web-1 | 2026/04/21 00:40:35 [notice] 1#1: start worker process 31
web-1 | 2026/04/21 00:40:35 [notice] 1#1: start worker process 32
web-1 | 172.21.0.1 -- [21/Apr/2026:00:40:44 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.19.0" "-"
```

5.4. Bajar servicios

```
~/Projects/docker-test 19:41
> docker compose down
[+] down 2/2
✓ Container docker-test-web-1 Removed
✓ Network docker-test_default Removed
```

5.5. Escalar servicio

```
~/Projects/docker-test 19:43
> docker compose up -d --scale web=3
[+] up 3/4
✓ Network docker-test_default Created 0.0
✓ Container docker-test-web-3 Starting 0.4
✓ Container docker-test-web-2 Started 0.2
✓ Container docker-test-web-1 Created 0.0
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint docker-test-web-3 (b1d5db854756b2fd705220905f505de33e00d7e2f42a323ab1879c915dc04b8): Bind for 0.0.0.0:8082 failed: port is already allocated
```

6. Dockerfile: Probar instrucciones principales

6.1. Ejemplo corto para probar todo

```
~/Projects/docker-test/dockerfile-test 19:47
> cat dockerfile

File: dockerfile
1 FROM alpine:3.2
2 LABEL autor="Test"
3 ARG VERSION=1.0
4 ENV APP_ENV=dev
5 WORKDIR /app
6 COPY archivo.txt /app/archivo.txt
7
8 RUN echo "version=$VERSION" > /app/version.txt
9
10 ADD archivo.txt /app/archivo2.txt
11
12 VOLUME ["/data"]
13
14 CMD ["sh", "-c", "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"]
```

```
~/Projects/docker-test/dockerfile-test 19:55
> docker build -t dockerfile --build-arg VERSION=2.0 .
[+] Building 1.2s (10/10) FINISHED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 345B
=> [internal] load metadata for docker.io/library/alpine:3.2
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/alpine:3.2
=> [internal] load build context
=> => transferring context: 52B
=> [2/5] WORKDIR /app
=> [3/5] COPY archivo.txt /app/archivo.txt
=> [4/5] RUN echo "version=2.0" > /app/version.txt
=> [5/5] ADD archivo.txt /app/archivo2.txt
=> exporting to image
=> => exporting layers
=> => writing image sha256:00deb4b83e59cac52473d607f138bd560e60ddc5cb0051b2d1f5e1d69d60a1e2
=> => naming to docker.io/library/dockerfile
```

```
~/Projects/docker-test/dockerfile-test 19:56
> docker run --rm dockerfile
APP_ENV=dev
archivo local
version=2.0
```


6.2. Inspeccionar label, env y volume

```
STDIN
1 [
2   {
3     "Id": "sha256:00deb4b83e59cac52473d607f138bd560e60ddc5cb0051b2d1f5e1d69d60a1e2",
4     "RepoTags": [
5       "dockerfile:latest"
6     ],
7     "RepoDigests": [],
8     "Comment": "buildkit.dockerfile.v0",
9     "Created": "2026-04-20T19:55:31.338165093-05:00",
10    "Config": {
11      "Env": [
12        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
13        "APP_ENV=dev"
14      ],
15      "Cmd": [
16        "sh",
17        "-c",
18        "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"
19      ],
20      "Volumes": {
21        "/data": {}
22      },
23      "WorkingDir": "/app",
24      "Labels": {
25        "autor": "Test"
26      },
27      "ArgsEscaped": true
28    },
29    "Architecture": "amd64",
30    "Os": "linux",
31    "Size": 5264544,
32    "GraphDriver": {
33      "Data": {
34        "LowerDir": "/var/lib/docker/overlay2/jcpyzuq3najzjmegepay7310f/diff:/var/lib/docker/overlay2/ijydScsdsdxr5z3otl7i9uui/diff:/var/lib/docker/overla
y2/g8j1j3pu6avkib3dye1swkp52/diff:/var/lib/docker/overlay2/aae00d845b5320585edac1f15e9f9da81b8317316f716d292582dd3ae79d141b/diff",
35        "MergedDir": "/var/lib/docker/overlay2/y945jr7cp2ttrzq1fw3iv13s/merged",
36        "UpperDir": "/var/lib/docker/overlay2/y945jr7cp2ttrzq1fw3iv13s/diff",
37        "WorkDir": "/var/lib/docker/overlay2/y945jr7cp2ttrzq1fw3iv13s/work"
38      },
39      "Name": "overlay2"
40    },
41    "RootFS": {
42      "Type": "layers",
43      "Layers": [
44        "sha256:2f0b1957d1f7074296e0d6388139b7a968e8c051f8b6227f3610757f7407af05",
45        "sha256:6406d58dc53346d4be66d43e1a792d549ed1f558d9f8c15c67f1c10f4c205383",
46        "sha256:e6f5eejc0f56c867ed64701be6f857c57f099fefe0ef4b0475fd80a03abe83eb",
47        "sha256:10b8278cc430299a3d458cfeb51da123f4ab6d67612cb3e61ee8245f979ad2ae",
48        "sha256:938419e4a838cf59875427aed3f4f3a09837b46d126aa0c9a857734709d4faf3"
49      ]
50    },
51    "Metadata": {
52      "LastTagTime": "2026-04-20T19:55:32.037668456-05:00"
53    }
54  }
55 ]
```